

ESTRUTURAS DE LEWIS

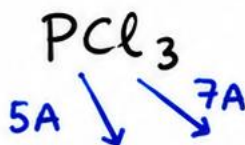
REPRESENTE AS ESTRUTURAS DE LEWIS DAS MOLÉCULAS A SEGUIR:



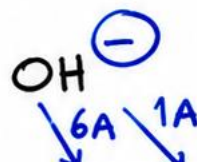
- ① CONTAR NÚMERO DE ELÉTRONS DE VALÊNCIA DA MOLÉCULA (N_{ev}). SE FOR CÁTION SUBTRAIR ELÉTRONS DE ACORDO COM A CARGA. SE FOR ÂNION, ADICIONAR e^- .



$$N_{ev} = 1 \cdot 1 + 7 \cdot 1 = 8$$

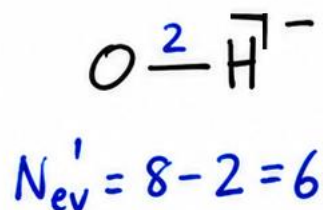
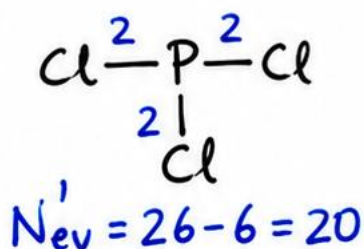
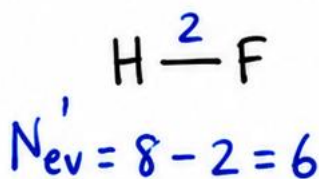


$$N_{ev} = 5 \cdot 1 + 7 \cdot 3 = 26$$

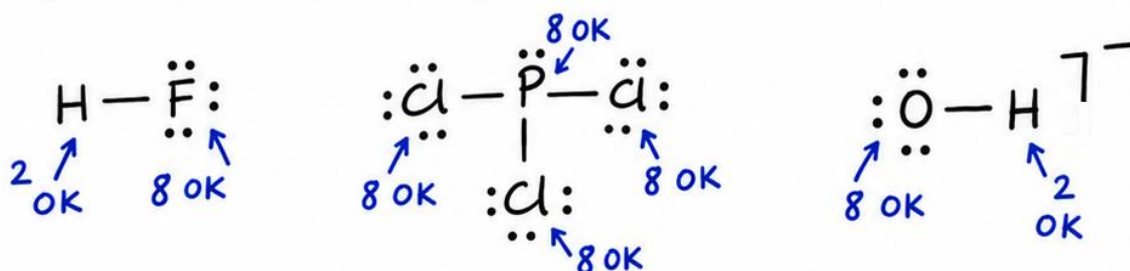


$$N_{ev} = 6 \cdot 1 + 1 \cdot 1 + 1 = 8$$

- ② CONECTAR ÁTOMOS POR LIGAÇÕES SIMPLES. SE ÁTOMOS ESTÃO EM MENOR QUANTIDADE, FAZEM MAIS LIGAÇÕES (ÁTOMOS CENTRAIS). RECONTAR ELÉTRONS DE VALÊNCIA, SUBTRAINDO OS USADOS NAS LIGAÇÕES.

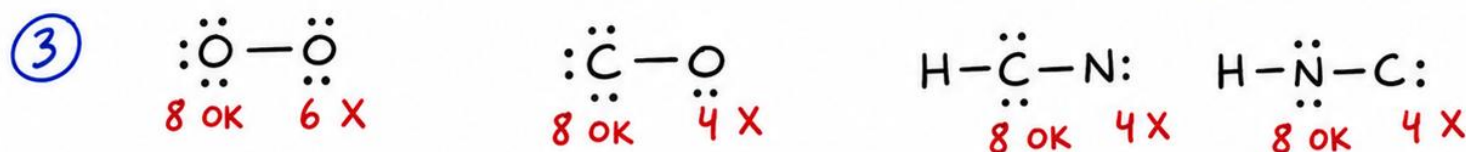
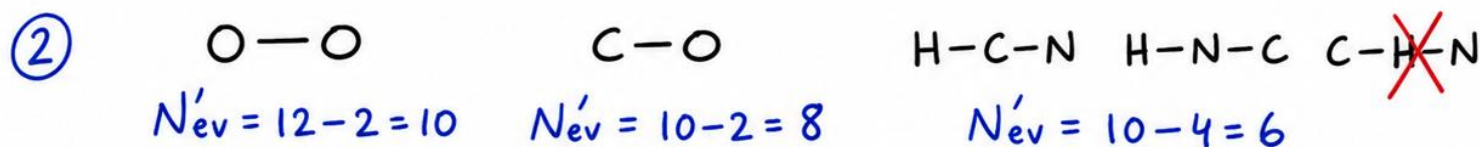
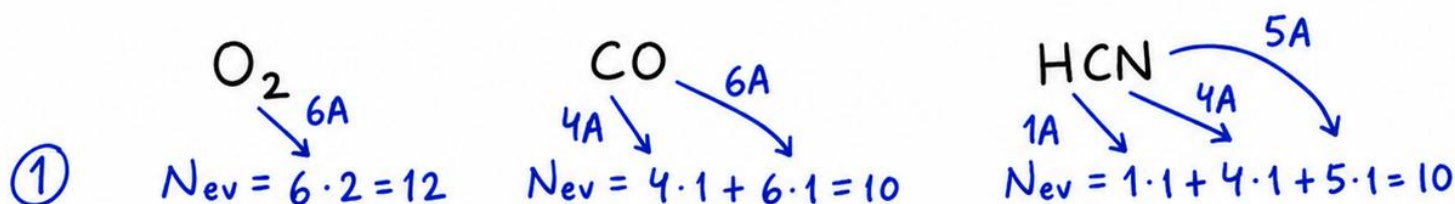


③ COLOQUE ELÉTRONS RESTANTES NAS CAMADAS DE VALÊNCIA DOS ÁTOMOS, DE MODO A PREENCHÊ-LAS:

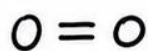


SE CAMADAS SÃO PREENCHIDAS, PARE AQUI

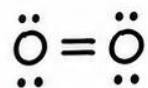
FAÇA O MESMO COM AS MOLÉCULAS A SEGUIR.



④ SE LIGAÇÕES SIMPLES NÃO LEVAM A CAMADAS COMPLETAS, FAZER LIGAÇÕES DUPLAS, RECONTAR ELÉTRONS RESTANTES E DISTRIBUA. SE OCORRER O MESMO, FAÇA TRIPLAS.

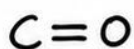


$$Ne'v = 12 - 4 = 8$$

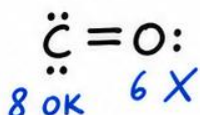


8 OK 8 OK

FINAL



$$Ne'v = 10 - 4 = 6$$



8 OK 6 X



$$Ne'v = 10 - 6 = 4$$

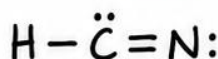


8 OK 8 OK

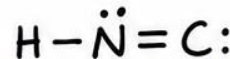
FINAL



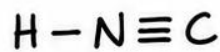
$$Ne'v = 10 - 6 = 4$$



2 OK 8 OK 6 X



2 OK 8 OK 6 X



$$Ne'v = 10 - 8 = 2$$



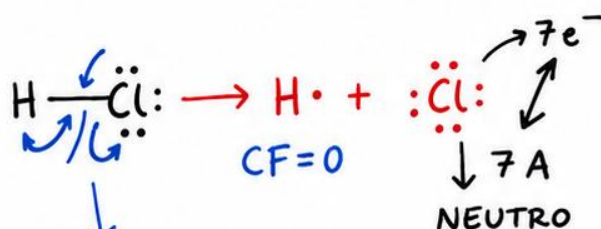
2 OK 8 OK 8 OK



2 OK 8 OK 8 OK

- 5) SE MAIS DE UMA ESTRUTURA É POSSÍVEL, CALCULAR CARGAS FORMAIS DOS ÁTOMOS. ESTRUTURA COM MENORES CARGAS É MAIS ESTÁVEL.

- 1) O QUE É: É CARGA DOS ÁTOMOS APÓS ROMPIMENTO DAS LIGAÇÕES E CADA ÁTOMO FICA COM $1e^-$ POR LIGAÇÃO



CF=0

7 A NEUTRO

CARGA TOTAL?

$$CARGA(HCl) = CF(H) + CF(Cl)$$

$$0 = 0 + CF(Cl)$$

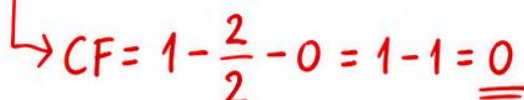
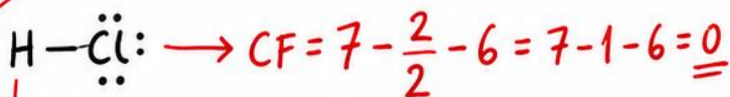
$$CF(Cl) = 0$$

- 2) COMO CALCULAR:

$$CARGA \text{ FORMAL} = \text{NÚMERO DE ELÉTRONS DE VALÊNCIA} - \frac{\text{NÚMERO DE ELÉTRONS LIGADOS}}{2} - \text{Nº DE ELÉTRONS LIVRES}$$

$$CF = V - \frac{B}{2} - L$$

FAMÍLIA

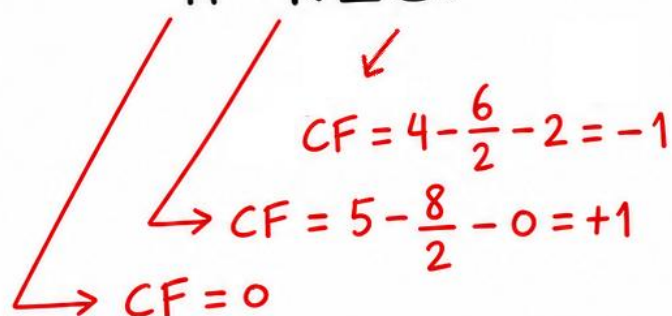
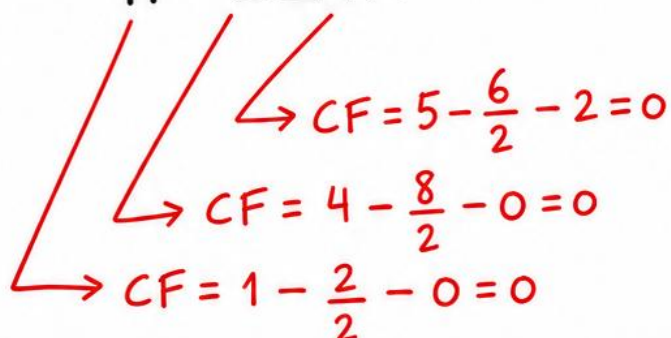


ESTRUTURAS DE LEWIS

$$CF = V - \frac{B}{2} - L$$



vs.



vs.

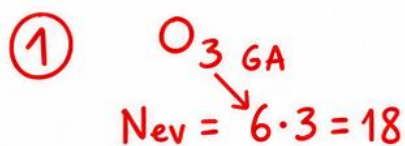


0 0 0

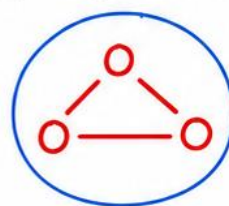
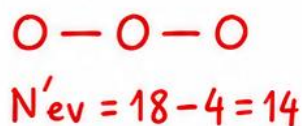
0 +1 -1

FINAL

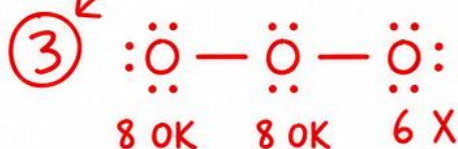
DETERMINE ESTRUTURA DE LEWIS DO O_3 :



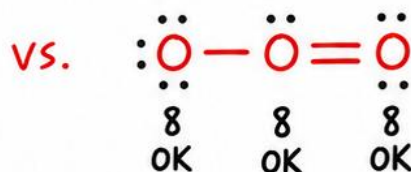
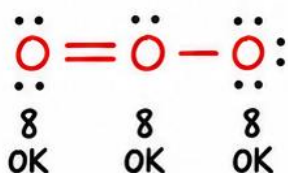
$\Rightarrow$ ②

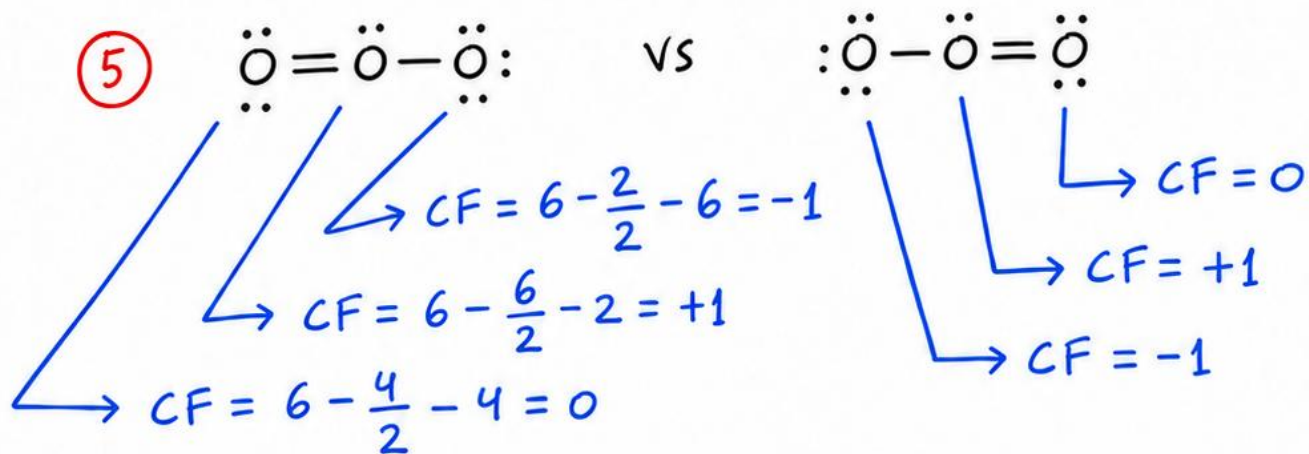


X IMPROVÁVEL POR GEOMETRIA



$\Rightarrow$ ④





⑥ SE MAIS DE UMA ESTRUTURA POSSUI CARGAS FORMAIS EQUIVALENTES, E SE DIFERENCIA PELA POSIÇÃO DA DUPLA/TRIPLA, TODAS SÃO VÁLIDAS. SÃO HÍBRIDOS DE RESSONÂNCIA: AUMENTAM ESTABILIDADE

